

Facial Emotion Recognition: Traditional ML vs. Deep CNNs

Introduction & Motivation

Facial expression recognition is a core problem in computer vision and human-computer interaction. The ability for machines to automatically detect human emotions has applications in areas such as mental health monitoring, user experience design, surveillance, and assistive technologies. Recent advances in machine learning, particularly deep learning, have significantly improved performance on image-based classification tasks, including emotion recognition.

However, building accurate emotion classifiers remains challenging due to variability in facial expressions, lighting conditions, and the subjective nature of emotions. Additionally, simpler machine learning models often struggle to capture the spatial relationships inherent in image data. This motivates a comparison between classical machine learning approaches and more modern deep learning techniques.

The goal of this project is to explore how different machine learning methods perform on the task of emotion classification from facial images. In particular, we aim to understand how feature representation (raw pixels vs. engineered features vs. learned features) impacts predictive performance.

Related Works

The evolution of Facial Emotion Recognition (FER) is defined by a transition from manual feature selection to automated hierarchical learning. While traditional algorithms required a separate, manual feature extraction phase (using techniques like Local Binary Patterns (LBP) or Principal Component Analysis (PCA)), Pranav et al. (2020) argue that the success of modern FER lies in the ability of Deep Convolutional Neural Networks (DCNN) to unify these steps. By utilizing multiple convolutional and pooling layers, DCNNs extract intricate facial features, allowing for high-accuracy classification across various emotions.

Furthermore, researchers have addressed the high data requirements of deep learning through Transfer Learning (TL). Akhand et al. (2021) demonstrate that pre-trained architectures, such as VGG-16 and ResNet-50, can be repurposed for emotion detection by leveraging previously

"learned" geometric shapes and edges from massive general datasets. This approach allows models to be fine-tuned on smaller, specialized facial datasets with minimal computational overhead. Beyond efficiency, these deep architectures offer superior "viewpoint invariance," providing the robustness necessary to handle the profile or angled views often found in real-world applications. By examining these advanced architectures alongside traditional methods, this project evaluates the threshold at which learned features begin to outperform engineered features.

Research Questions

This project is guided by the following research questions:

1. How well can classical machine learning models (e.g., Logistic Regression, K-Nearest Neighbors) classify facial emotions using raw pixel data?
2. Does feature engineering (e.g., Histogram of Oriented Gradients) improve classification performance compared to raw pixel inputs?
3. How does a Convolutional Neural Network (CNN) compare to classical approaches in terms of accuracy and robustness?

We expect that:

- Models using raw pixel data will perform the worst due to a lack of structure.
- Feature engineering will improve performance.
- CNNs will outperform both by learning hierarchical image features.
- Using class weights will help the model learn faster and help improve performance

Data Description

The dataset used in this project consists of labeled facial images categorized into discrete emotion classes, including anger, happiness, sadness, surprise, fear, and more. The data was obtained from an online repository and organized into folders by emotion class. These pictures were classified by prompting the person to exaggerate the emotion they are feeling for the picture, so the model should be able to find features in each emotion to differentiate between them.



Anger and happiness, respectively

Structure of the Data

- Each observation is a grayscale image of a face.
- Images are resized to 48×48 pixels.--> 2304 features for raw pixels
- Pixel values range from 0 to 255 and are normalized to the range $[0, 1]$.
- The dataset contains multiple emotion categories: angry (890 images), contempt (208 images), disgust (239 images), fear (570 images), happiness (1406 images), neutrality (524), sadness (746 images), & surprise (775 images).

Variables

- Input (X): Image pixel values (either flattened or structured as 2D arrays)
- Output (y_{hat}): Emotion label (categorical variable)
- Actual(y): Actual emotion classified to the input

Data Preparation

Several preprocessing steps were applied:

- Conversion to grayscale (if not already)
- Resizing all images to a fixed dimension (48×48)
- Normalization of pixel intensities
- Encoding of categorical emotion labels into numeric values
- Splitting into training and testing sets (80/20 split)

Additionally, class imbalance was addressed using class weighting techniques to ensure that underrepresented emotions were not ignored during training.

Methods

To address the research questions, we implemented and compared three types of models:

1. Logistic Regression (Raw Pixels)

Images were flattened into 1D vectors and used as input features. Logistic Regression attempts to learn linear decision boundaries between emotion classes. We plan to use this logistic regression as a baseline evaluator to compare a linear model to more complex data analysis methods.

2. Feature Engineering with HOG + Classical Models

We used the Histogram of Oriented Gradients (HOG) to extract meaningful features from images. We are capturing the edge directions and local shape information, which is important for facial structure, and thus determining emotion. We can then use these features we are extracting to run different models, such as a logistic regression or a K-Nearest Neighbors.

3. Convolutional Neural Network (CNN)

The model was implemented using the PyTorch framework, following a standard deep learning pipeline for image classification. Its architecture is composed of several specialized layers designed to process visual data: convolutional layers are employed to perform automated feature extraction, followed by pooling layers that provide dimensionality reduction to maintain computational efficiency. Finally, fully connected layers integrate these extracted features to perform the high-level classification of facial emotions.

By using a CNN, we can automatically learn about more complex feature patterns and have a stronger understanding of the image data. The CNN approach would be able to extract the features automatically and learn directly from the data.

We ran our CNN using three layers and with 30-50 epochs, occasionally stopping when the learning threshold is met, and there are diminishing returns for the epochs. When running the model with class weights, we boosted the strength of guessing for contempt, disgust, and neutral, as they were rarely guessed before being added, and when removing the weights again, we increased the number of epochs to compensate for the lack of weights.

Evaluating the Accuracy of Models

To evaluate the model's performance, several key metrics were utilized. Accuracy was calculated to determine the overall proportion of correct predictions relative to the total number of samples. To gain a more granular understanding of model errors, a confusion matrix was employed, which reveals specific patterns of misclassification and identifies which emotions are being confused with one another. Furthermore, precision, recall, and the F1-score were analyzed to assess the model's handling of false positives and false negatives, providing a balanced measure of its predictive reliability across different emotional categories.

Results

Classical Models (Raw Pixels)

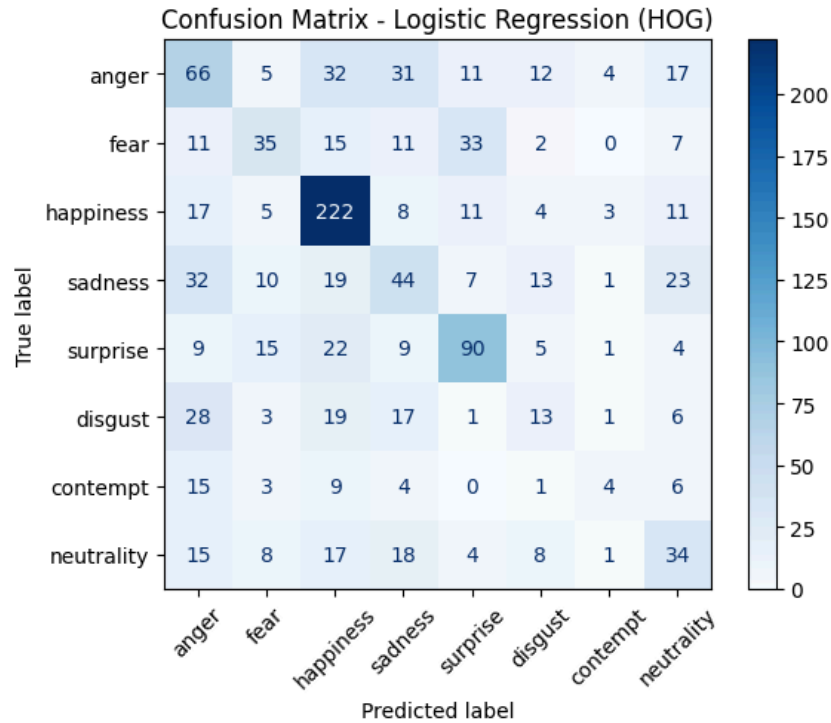
Logistic Regression using raw pixel values produced a relatively low accuracy. The low accuracy of 36% is telling us that the logistic regression model was able to correctly classify the emotion 36 out of 100 times.

When using the raw pixels with the KNN model, we saw our test accuracy reduce to 27%, or only correctly classifying the emotion 27 out of 100 times. Since KNN models rely on the distance between points to classify them, the high dimensions of the data make the calculations for distance harder and unreliable, leading to a lower accuracy. Since the model is taking in 2304 features, the logistic regression can learn patterns across the data set, while the KNN struggles with the distance between points.

HOG Features + Classical Models

Models trained on HOG features showed noticeable improvement over raw pixel inputs. Due to the feature extraction being more compact and concise, the logistic and the KNN models are able to benefit from the new features and increase their accuracy in correctly identifying emotions.

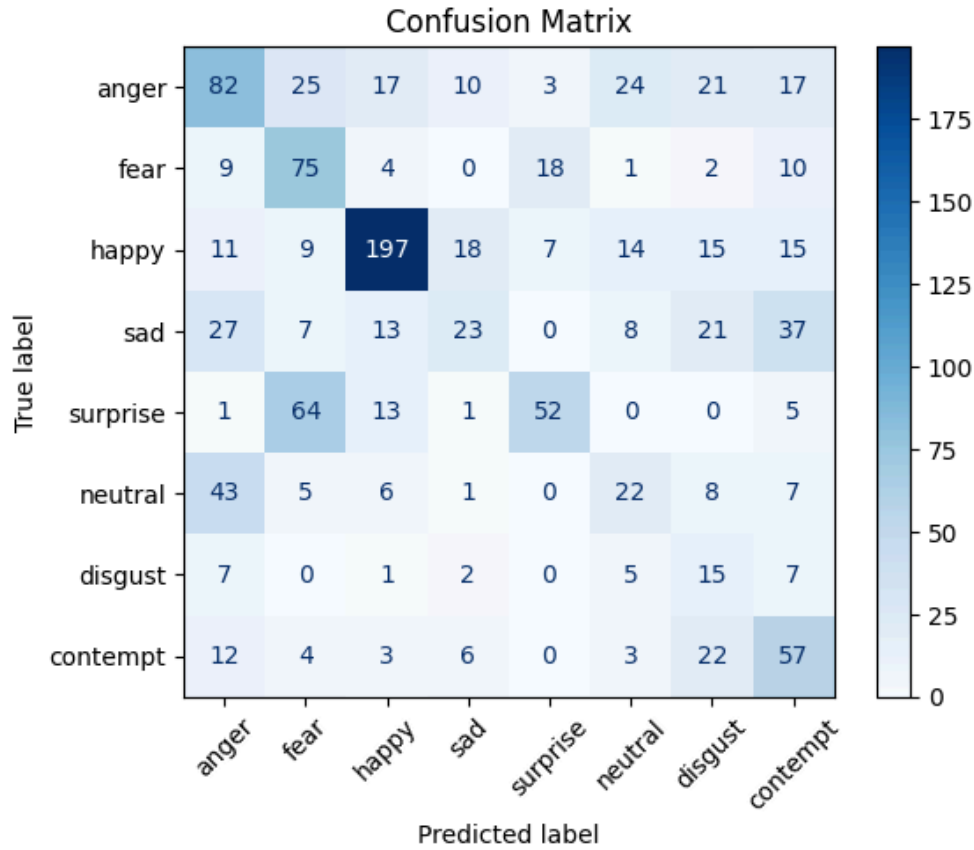
When using the HOG feature extractions, we saw the classification test accuracy increase from 36% to 44% with logistic regression, and from 27% to 42% for KNN models. With less complex features and a lower dimensionality, the KNN model is able to perform better, but there is still some complexity to the distance between points, as the logistic regression continues to outperform it.



Confusion Matrix of the Logistic Regression with HOG Feature Extraction

CNN Model

The CNN model did not perform as well as expected in this study. While CNNs are theoretically the most powerful approach for image classification, this specific implementation faced several challenges, including limited training iterations, the potential lack of a proper training loop, and insufficient data or hyperparameter tuning. As a result, the CNN did not significantly outperform the classical models in its current state. In fact, it proved less accurate than the other models, struggling to identify specific emotions, such as happiness, with consistent precision. Compared to the 36% accuracy achieved by previous methods, the CNN model reached only 32% accuracy. Since a well-optimized CNN should outperform both the logistic regression and KNN models, future adjustments will focus on increasing the training duration to allow the model to converge on more accurate hyperparameters. Furthermore, incorporating batch training and refining the class weights will be essential to improving accuracy and ensuring the model performs according to theoretical expectations.



Confusion matrix for CNN model with 30 epochs, 3 layers, and no Class Weights

Error Analysis

Confusion matrices revealed that certain emotions were frequently misclassified, particularly those with similar visual features. We can see that for the logistic regression with HOG feature extraction, there is a higher likelihood that disgust is incorrectly labeled as anger than correctly labeled as disgust.

Discussion & Conclusion

The results of this study demonstrate that while raw pixel data is insufficient for strong performance in classical models, feature engineering techniques like HOG significantly improve outcomes by incorporating vital structural information. Currently, the logistic regression model with HOG extraction provides the most reliable classification; however, the iterative learning process of the CNN showed promise. Even without optimization, the CNN began identifying key

facial features that suggest it would eventually surpass classical methods given more refined training.

The observed accuracy—ranging from 36% for classical models to 32% for the initial CNN—is notably lower than state-of-the-art results, which can be attributed to several factors. First, classifying seven distinct emotions is mathematically complex; while 32–36% is low, it significantly outperforms a random-guess baseline of ~14%. Much of the error likely stems from inter-class similarity, where the subtle muscle movements distinguishing "Fear" from "Surprise" or "Anger" from "Disgust" create ambiguous decision boundaries. Furthermore, while established research by Akhand et al. (2021) relies on Transfer Learning and massive datasets, this project's CNN was trained from scratch on a relatively small scale. Without a pre-trained starting point or tens of thousands of images, deep learning models are highly prone to underfitting.

Attempts to improve performance via class weighting for underrepresented emotions—such as disgust and contempt—actually resulted in a net decrease in accuracy. By overcompensating for these minority classes, the model's proficiency in identifying majority classes like "Anger" was reduced. This highlights a critical trade-off in model tuning: balancing sensitivity toward rare classes without sacrificing overall predictive power. Future work will focus on fine-tuning hyperparameters, exploring alternative hand-crafted features, and implementing a more robust, optimized CNN architecture to bridge the gap between our current results and theoretical potential.

Limitations

We identified several constraints that impact the generalizability of this study. Primarily, the dataset size is likely insufficient for training a robust deep learning model from scratch, and inherent class imbalances continue to skew results despite weighting attempts. Beyond technical optimizations, the subjective nature of human emotion remains a fundamental hurdle. Our models may struggle to extrapolate to candid photography, where individuals often display complex, overlapping emotional states rather than the singular, distinct expressions found in controlled datasets.

Future Work

To improve the model's performance, future work will focus on refining the CNN training process through the systematic tuning of hyperparameters, including the number of epochs, optimizer selection, and learning rate adjustments. To address potential overfitting and expand the diversity of the training set, data augmentation techniques can be applied to the input images. Furthermore, the scope of the project can be broadened by experimenting with deeper neural architectures and leveraging the power of transfer learning by incorporating pre-trained models, which have already learned complex feature representations from larger, generalized datasets.

We would also be curious to look at a different data set in the future, with varying emotions, and analyze how the model classifies unfamiliar emotions. Since the model will most likely select the most similar emotion, it is useful to question how we can enforce the understanding of new emotions and correct classifications (reinforcement learning).

This project highlights the progression from simple machine learning models to more advanced deep learning approaches in image classification. While classical methods benefit from feature engineering, CNNs offer the most promise when properly implemented. This work provides a foundation for further exploration into emotion recognition and demonstrates key challenges in applying machine learning to real-world data.

References

Akhand, M. A. H., et al. "Facial Emotion Recognition Using Transfer Learning in the Deep CNN." *Electronics*, vol. 10, no. 1036, 27 Apr. 2021, pp. 1-19. *MDPI*, <https://doi.org/10.3390/electronics10091036>.

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